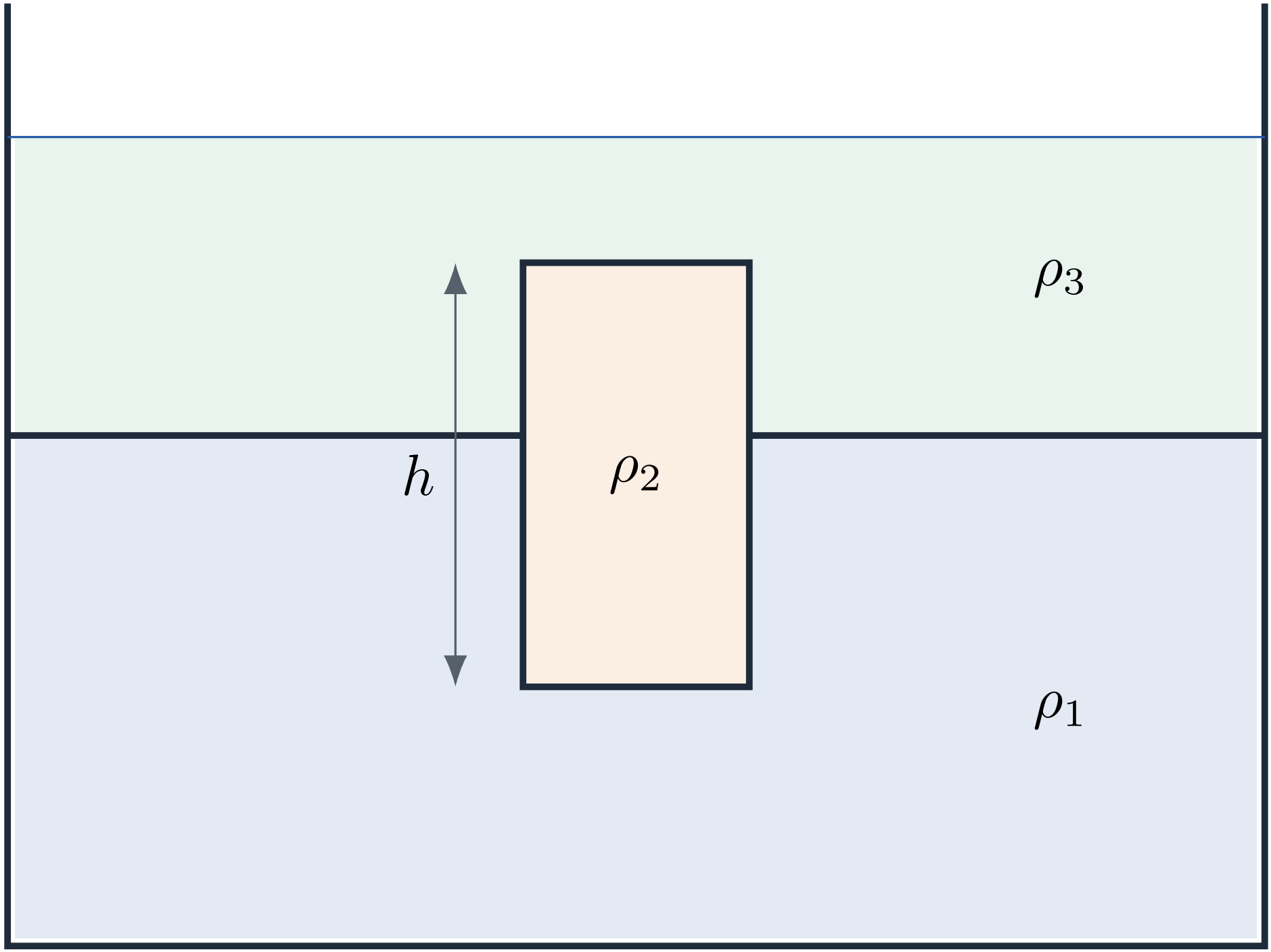
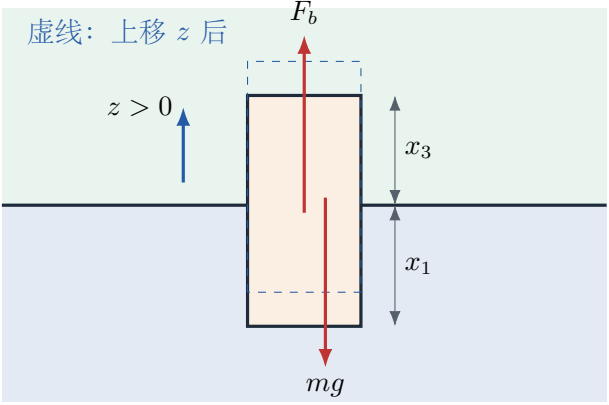


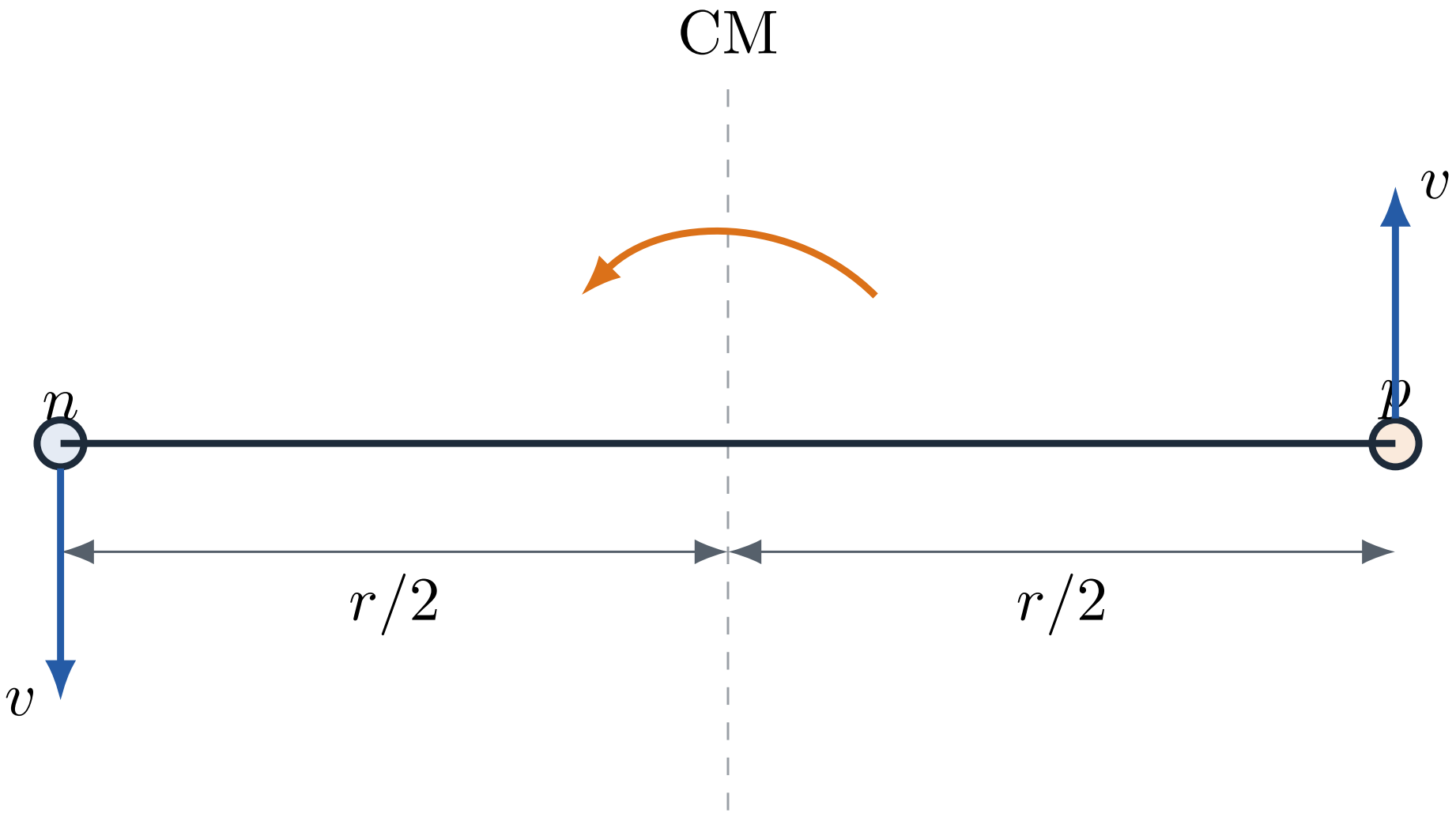
$\rho_1 > \rho_2 > \rho_3$，圆柱截面积为 S



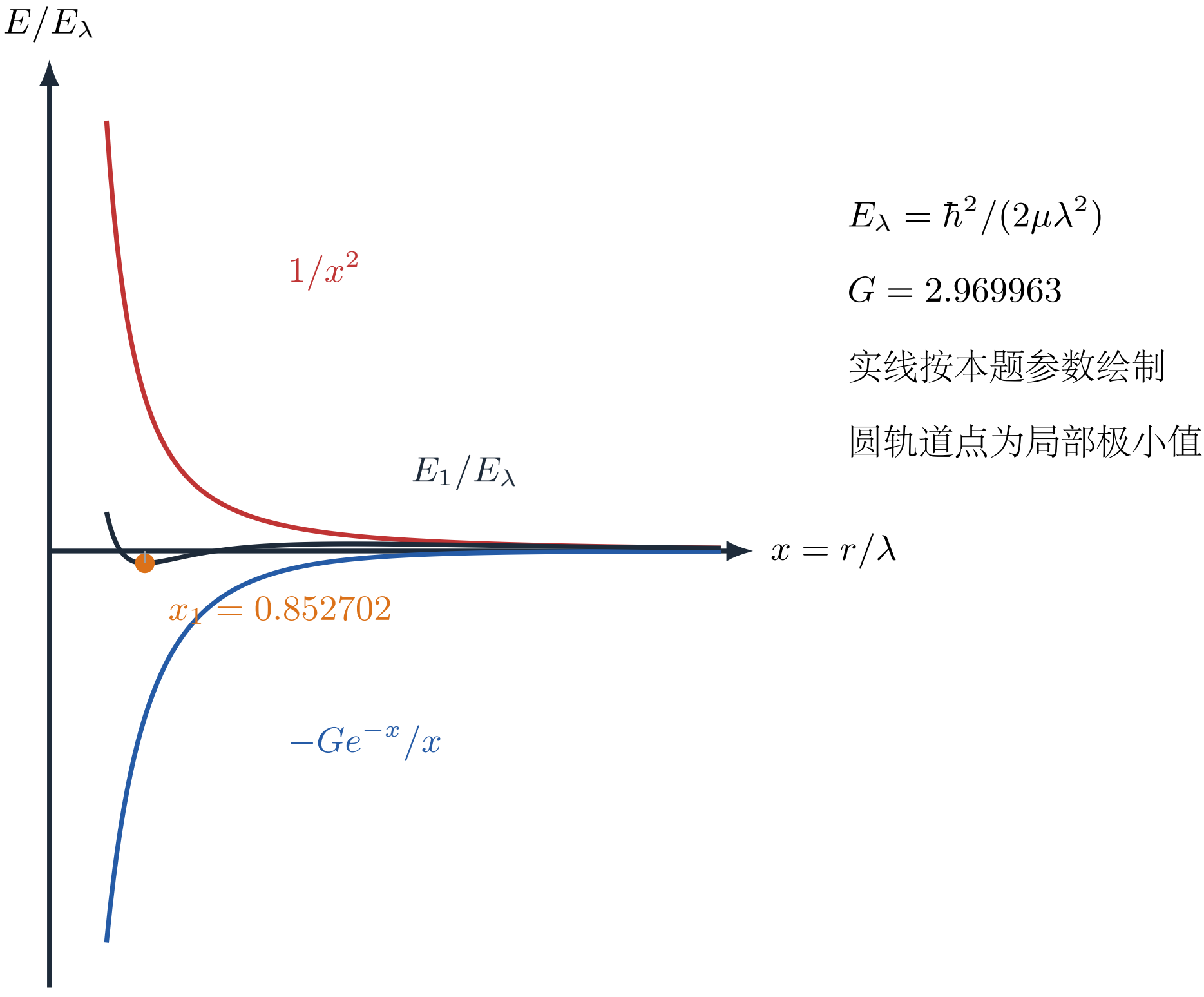
$$\rho_1 x_1 + \rho_3 x_3 = \rho_2 h$$

$$x_3 = h\frac{\rho_1 - \rho_2}{\rho_1 - \rho_3}$$

$$\Delta F_b = -Sg(\rho_1 - \rho_3)z$$



$$U(r) = -g_2^2 e^{-r/\lambda}/r\,, \;\; \lambda = \hbar/(m_\pi c)$$

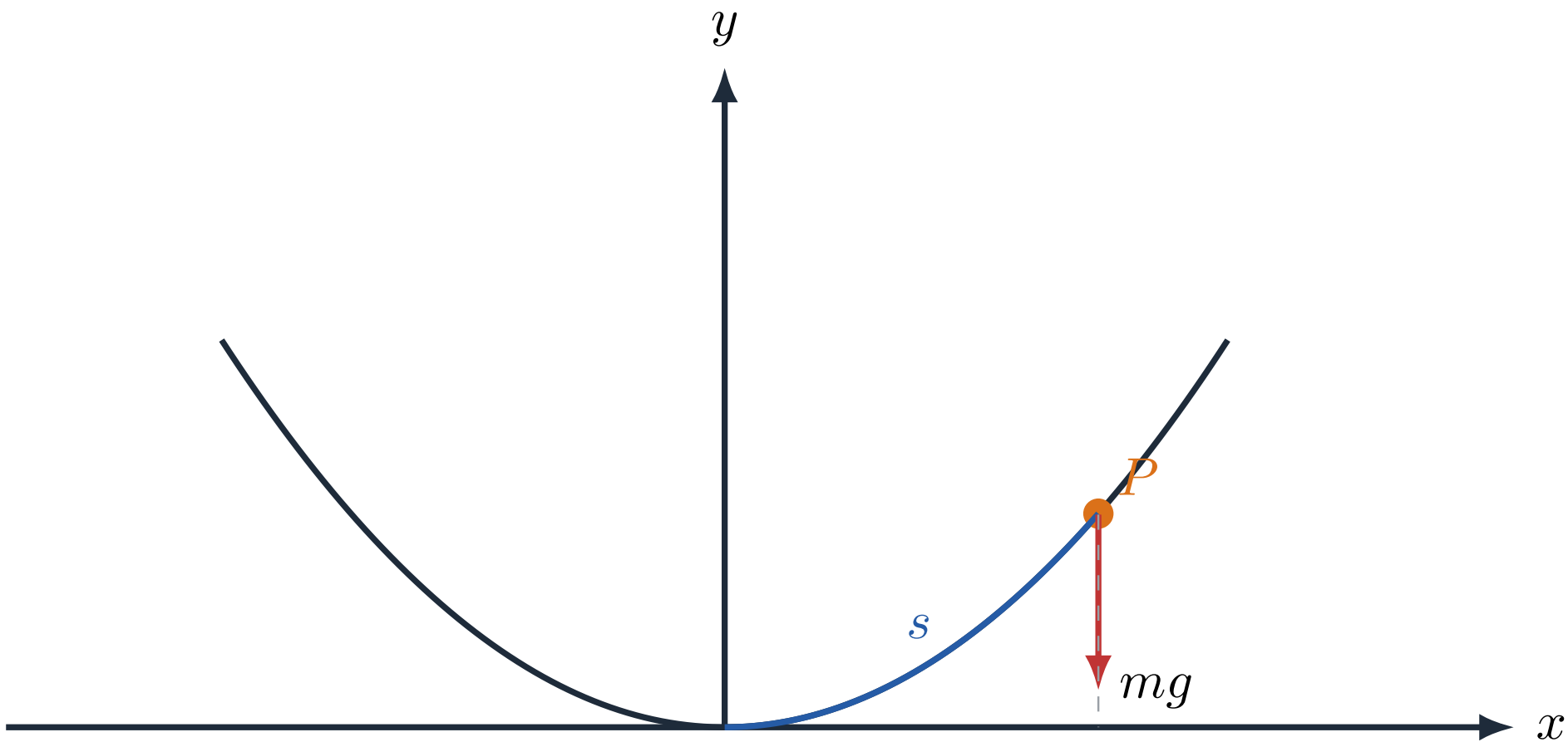


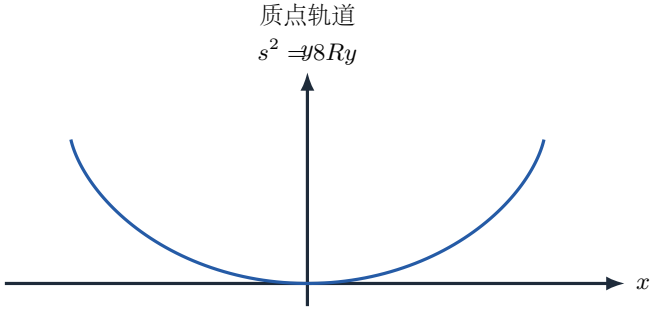
$E_\lambda = \hbar^2/(2\mu\lambda^2)$

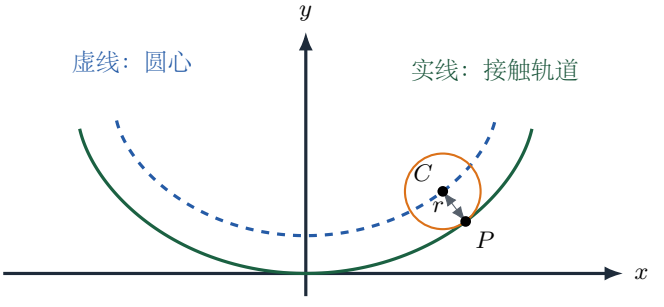
$G = 2.969963$

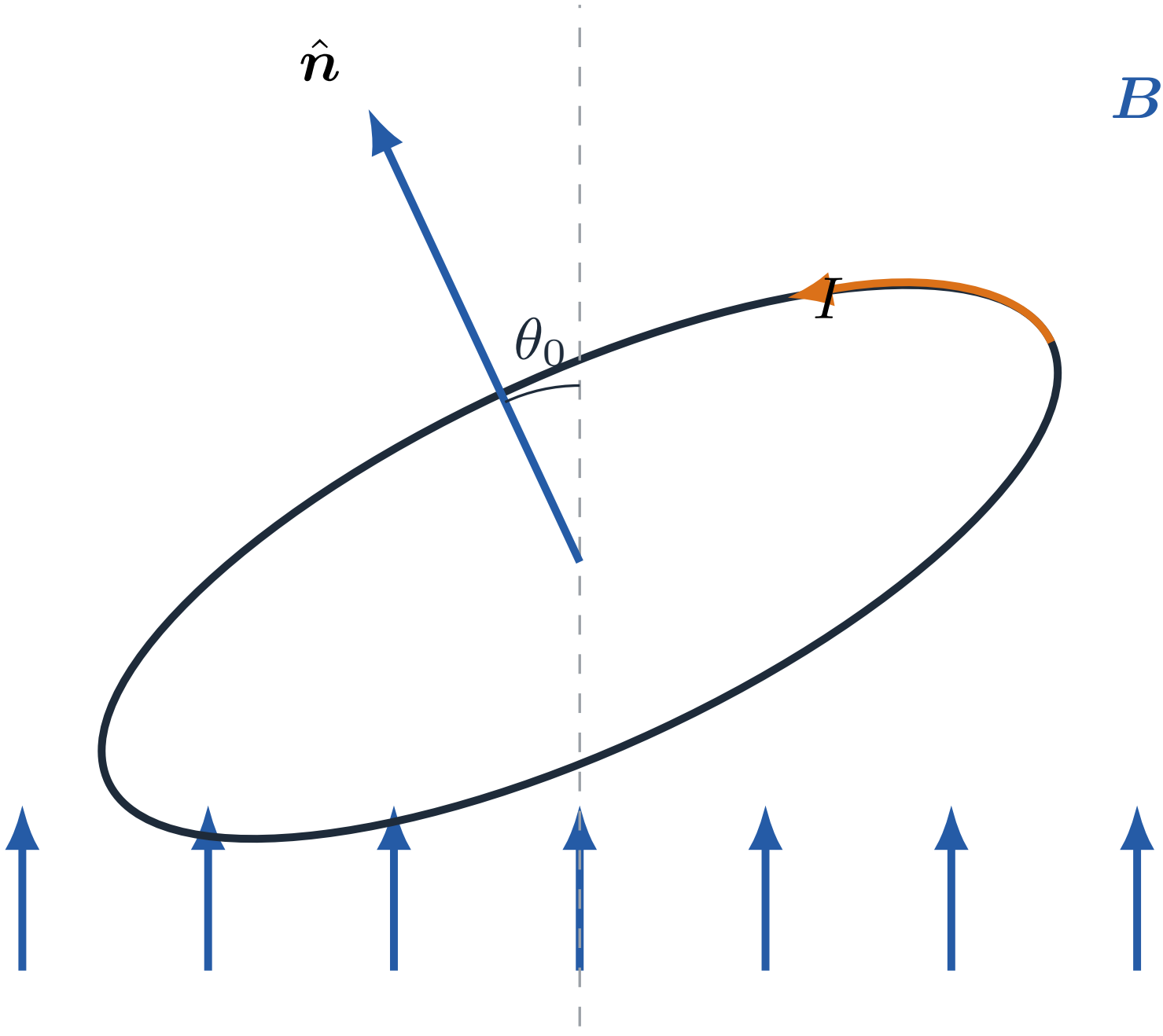
实线按本题参数绘制

圆轨道点为局部极小值


y
P
s
mg
x

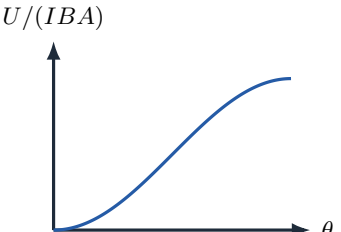

质点轨道
$s^2 = 8Ry$
x


y
虚线：圆心
实线：接触轨道
C
r
P
x



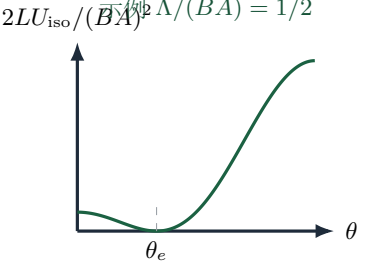
电流正向按 $\hat{n}$ 的右手定则；圆环为投影示意

恒流：$1-\cos\theta$



孤立环：$(\Lambda/(BA)-\cos\theta)^2$

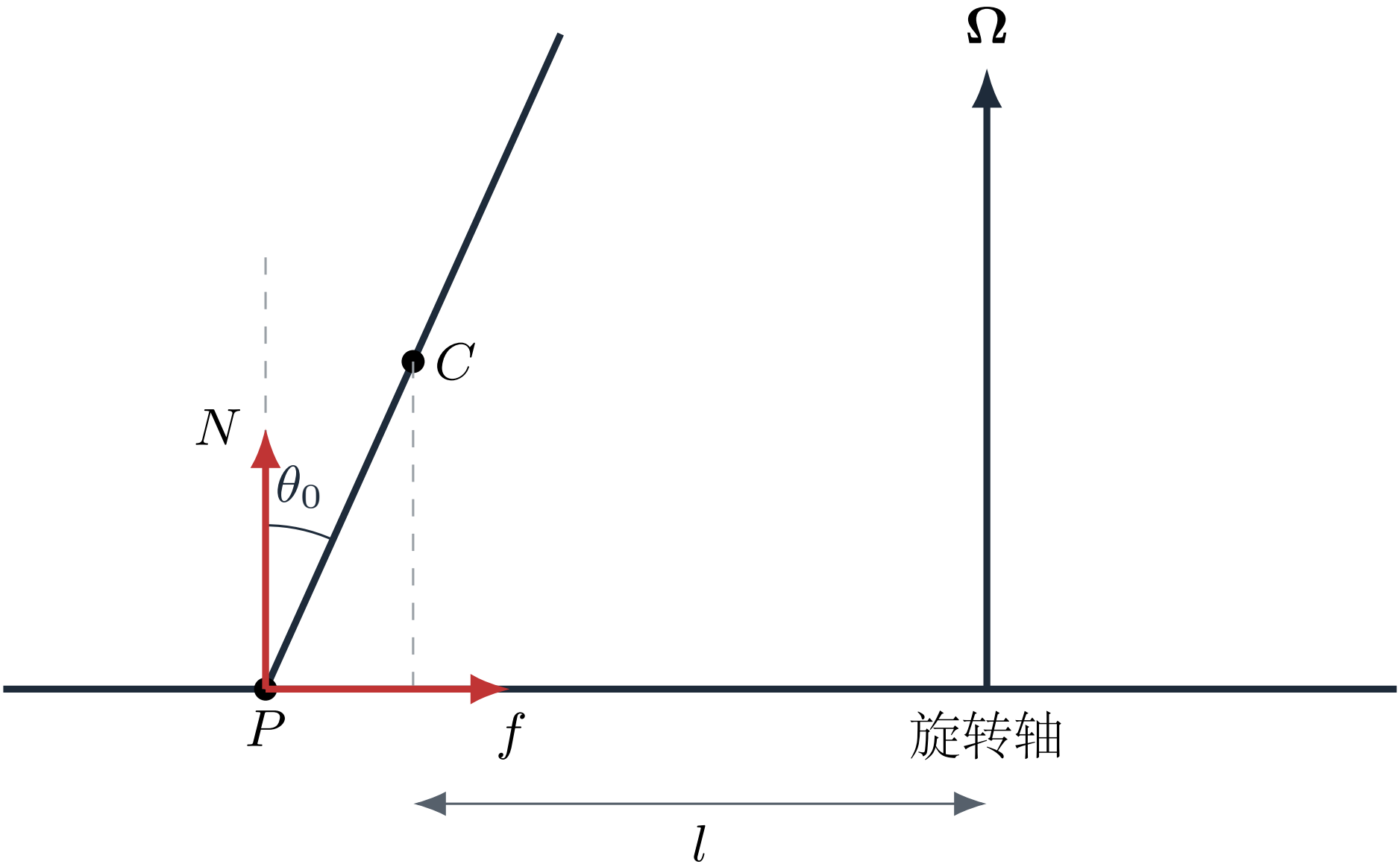
示例 $\Lambda/(BA)=1/2$



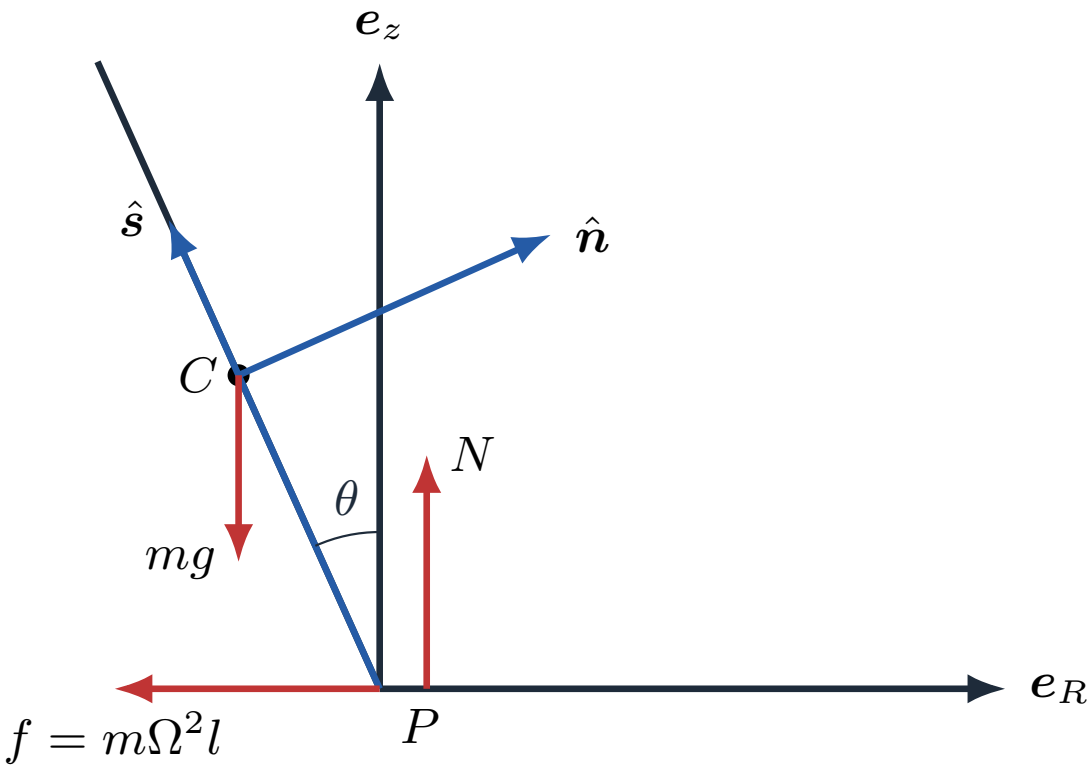
$$\Delta\Phi = BA(1-\cos\theta_0)$$

$$I_f = I - \Delta\Phi/L$$

$$K_f = I\Delta\Phi - \Delta\Phi^2/(2L)$$



径向—竖直截面；硬币以直径线表示，向旋转轴倾斜

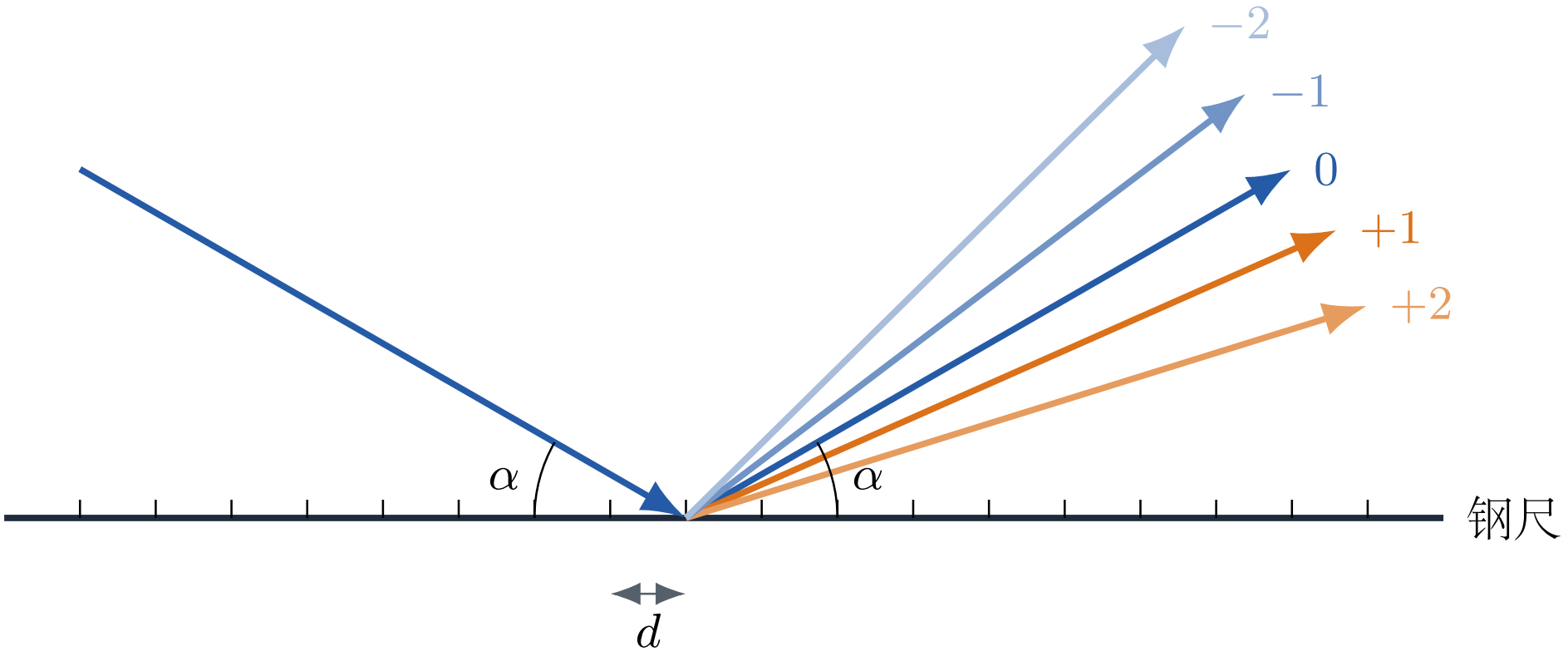


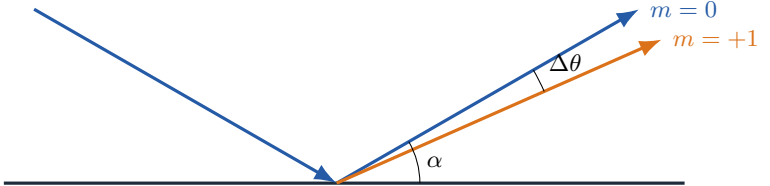
$$\omega_n = -\Omega l/r$$

$$\omega_x = -\Omega(l/r + \sin\theta)\cos\theta$$

$$\omega_y = \Omega(\cos^2\theta - l\sin\theta/r)$$

摩擦沿 $-\boldsymbol{e}_R$；接触力矩不为零

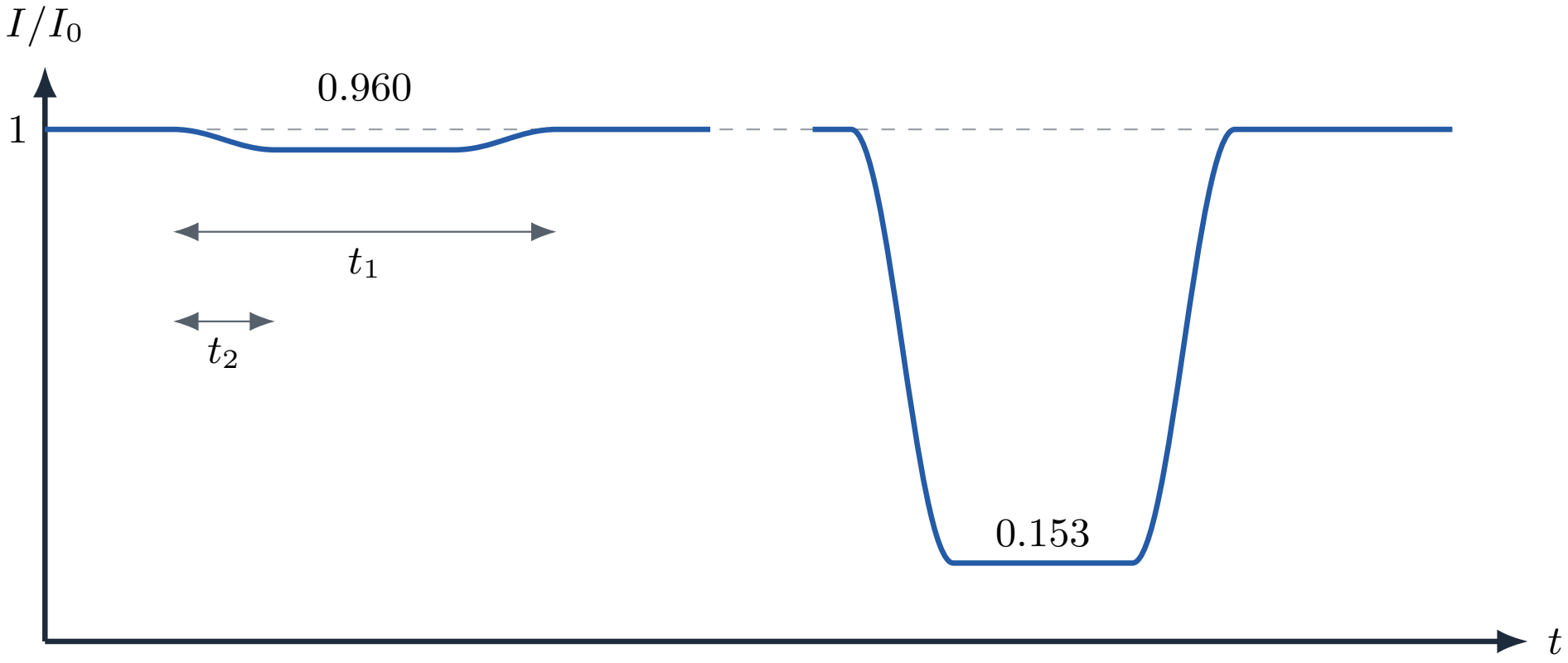

−2
−1
0
+1
+2
α
α
钢尺
d



$$d(\cos\beta_m - \cos\alpha) = m\lambda$$

$$\beta_{+1} < \alpha\,;\;\; \beta_{-1} > \alpha$$

$$\lambda \simeq d \sin\alpha\, \Delta\theta$$

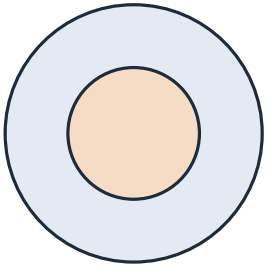


纵坐标按光度比例；两食间隔压缩，横向时长示意



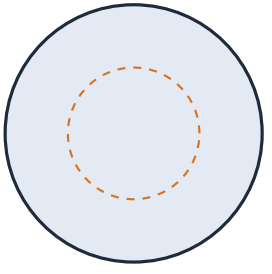
$a_B/a_A = K_2/K_1 = 12$

A, B 与亮暗组分的配对尚未给定



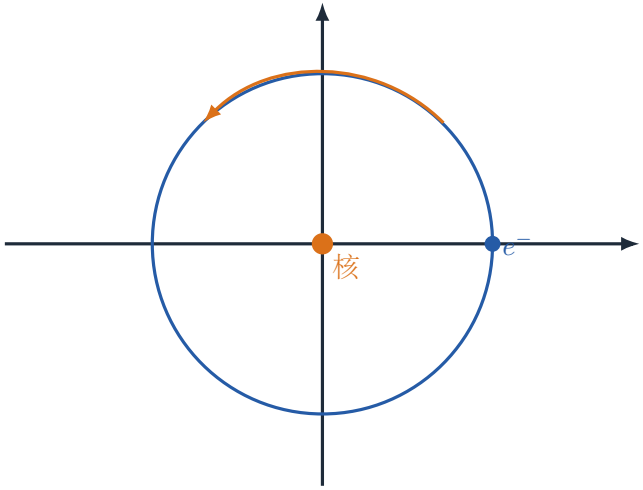
浅食：小亮星在前

损失暗星的一块投影

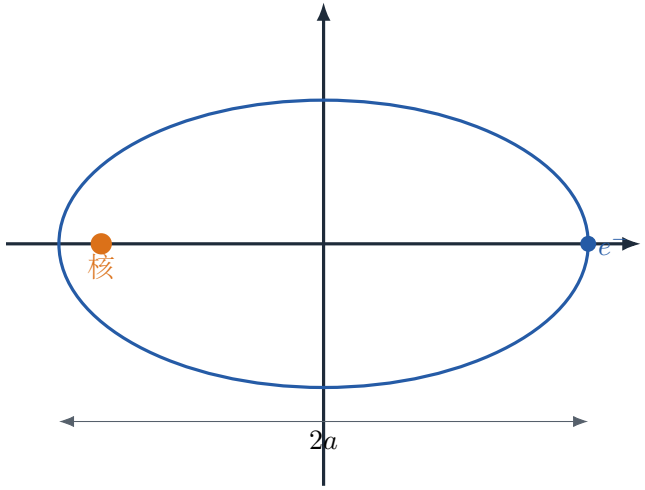


深食：小亮星在后

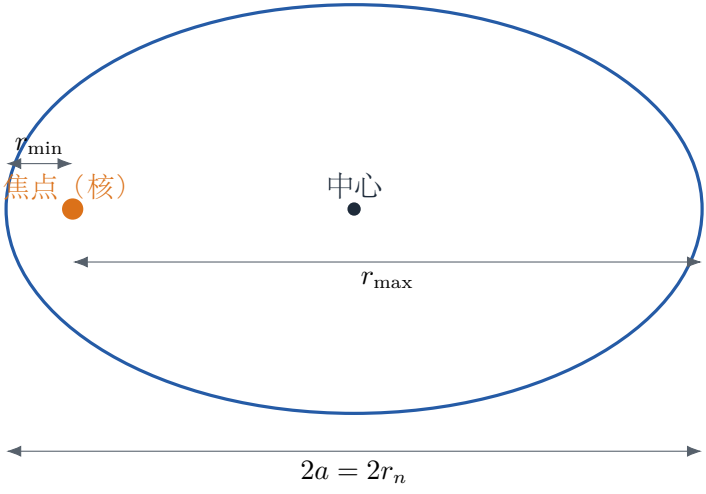
虚线为被完全遮挡轮廓



玻尔圆轨道



Sommerfeld 椭圆轨道



$$a = r_n = \frac{n^2\hbar^2}{m_e\kappa}$$

$$e = \sqrt{1 - \frac{l^2}{n^2}}$$

$$r_{\min} = a(1 - e)$$

$$r_{\max} = a(1 + e)$$